

Resonant MEMS Mirror and Phase Locked Loop

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Experiments

1.1 Hysteresis Points

After the startup procedure, explained in Section 3.3.1 we operated the system in open loop. In our testing, the system jumped to the upper part of the hysteresis curve at a frequency of

$$f_{\text{jumpUp}} = 4250\text{Hz} \quad (1.1)$$

when approaching on the lower part of the hysteresis curve from the right (decreasing frequency). A second jump from the upper to the lower response curve occurred at

$$f_{\text{jumpDown}} = 4620\text{Hz} \quad (1.2)$$

when approaching on the upper part of the hysteresis curve from the left (increasing frequency).

1.2 Maximum Mirror Angle

We measured the amplitude of the PSD voltage to be

$$V_{\text{diff,pp}} = 9.425 \text{ V} . \quad (1.3)$$

Using this, we can calculate the peak to peak mirror position

$$y_{\text{pp}} = \frac{L}{2} \frac{V_{\text{diff,pp}}}{V_{\text{sum}}} = \frac{30 \times 10^{-3}}{2} \frac{9.425\text{V}}{6.25\text{V}} = 0.04524 \quad (1.4)$$

. With this trigonometry and the distance from the mirror to the sensor

$$D_{\text{op}} = 0.035\text{m}, \quad (1.5)$$

this is converted into the maximum amplitude of the resulting mirror angle

$$\Theta_{\text{max}} = 0.5 \tan\left(\frac{y_{\text{pp}}}{2D_{\text{pp}}}\right)^{-1} = 0.5 \tan\left(\frac{0.04524}{0.07}\right)^{-1} \approx 0.28688 \approx 16.44^\circ \quad (1.6)$$

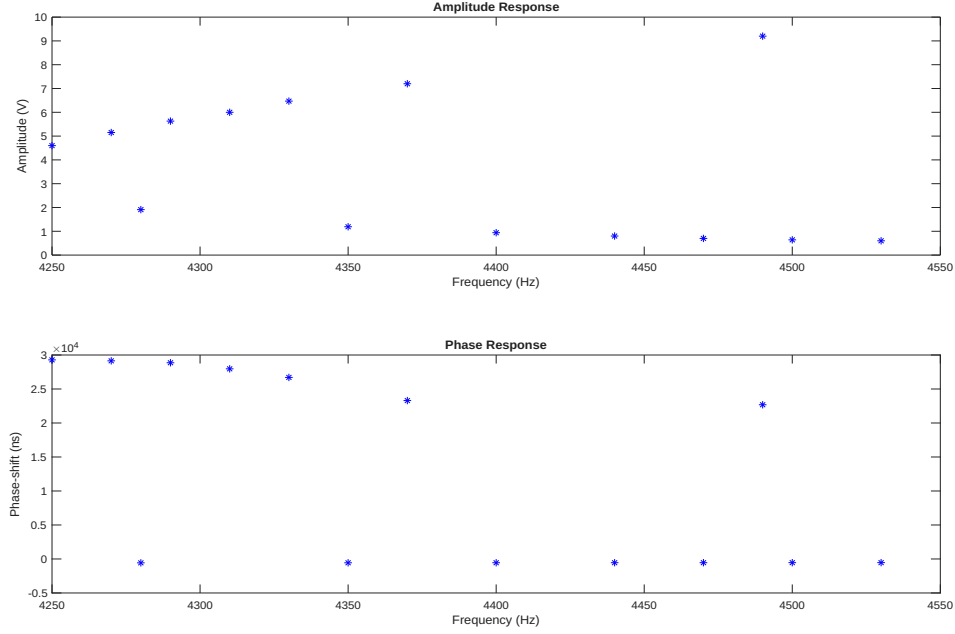


Figure 1.1: Measured frequency response of the system (1.1)

1.3 Frequency Response

At 14 different frequencies, V_{diff} , and the time delay was measured. The resulting points are plotted in Figure 1.1. In the amplitude response of the mirror, the hysteresis quality is clearly visible. Also, both the previously recorded jumps from lower to upper and to lower response curve from Section 1.1 further validate the recorded points.

The phase response also shows clear hysteresis, with slightly negative time delay on the lower response curve and clear positive time delay that decreases with increasing frequency on the upper response curve.

1.4 Zero Crossing of the Mirror

Operating the system at the upper right end of the hysteresis curve, we measured the slope of the zero crossing of V_{diff} to be

$$m_{V_{\text{diff}}} = \frac{3.5mV}{20\mu s} = 175 \frac{V}{s} \quad (1.7)$$

1.5 Noise Measurement

Using the oscilloscope, we measured the voltage standard deviation

$$\sigma_I = 43 \text{ mV}, \quad (1.8)$$

with which, we calculate the timing jitter

$$\sigma_{\Delta t} = \frac{1}{|m_{V_{\text{diff}}}|} \sigma_I = \frac{1}{175 \frac{\text{V}}{\text{s}}} 0.043 \text{ V} = 245 \text{ } \mu\text{s} \quad (1.9)$$

of our system.

1.6 Matlab Timing Jitter

Using the provided MATLAB function AC-RMS we measured the timing jitter again, which resulted in a much lower value:

$$\sigma_{\Delta t}^{\text{MATLAB}} = 18.85 \text{ } \mu\text{s} \quad (1.10)$$

The much lower value can be explained with the noise removal algorithm, that is used on the FPGA for measuring the timing jitter via the MATLAB script.

1.7 Full Shape Comp Drive Signal

While operating the system at the upper right end of the hysteresis curve (top response branch), we switched the mirror to IDLE mode to observe the natural decay of its oscillation. This procedure allows us to measure the displacement current—and the corresponding voltage—generated solely by the mirror’s motion, without active actuation. The captured signal is shown in Figure 1.2.

1.8 Setting Operating Point parameter

Because the system behaves approximately linearly in a small region around a chosen operating point, a linear control approach is appropriate. To implement this, the system is linearized around the operating point. Measurements are offset by subtracting the operating point values, so the controller operates on deviations from the nominal state. The controller then computes a corrective action based on these deviations. This output is added to the nominal control input associated with the operating point, resulting in the final control signal applied to the system. This method ensures that

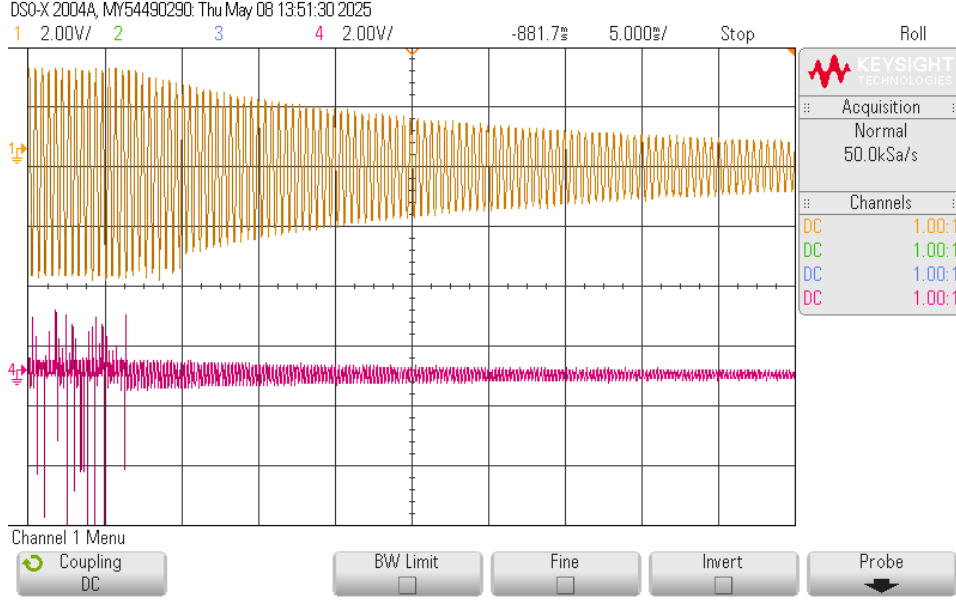


Figure 1.2: Decaying oscillation of the switched off mirror.

the linear controller remains effective within the valid range of the linear approximation. We adapted the offset values for our operating point:

$$f_{\text{lin}} = 4580 \text{ Hz}, \quad (1.11)$$

$$f_{\varphi, \text{lin}} = 16200 \text{ ns}. \quad (1.12)$$

1.9 Step Response and Model Identification

Following procedure 3.3.2, we obtained the phase response of the system to a small abrupt change in input frequency. After subtracting the operating point, we can estimate a continuous second order model of the system. 1.3 shows the estimated model over the recorded step response. Investigating the bode diagram 1.4, the continuously positive phase over the whole frequency response makes little sense for a real system. After investigation we found an error in our implementation of the homework. Fixing the error and doing the system estimation again, yielded the more sensible Bode plot shown in 1.6. The pole-zero map of this corrected system is also seen in 1.7. Because we only noticed the error after the lab was completed, we performed all further experiments with the ill-defined system. Fortunately we were still able to stabilize the system both with the pure integral and the PI controller, as we will demonstrate in the next sections. The given values in future experiments also stem from the calculations from the ill-defined

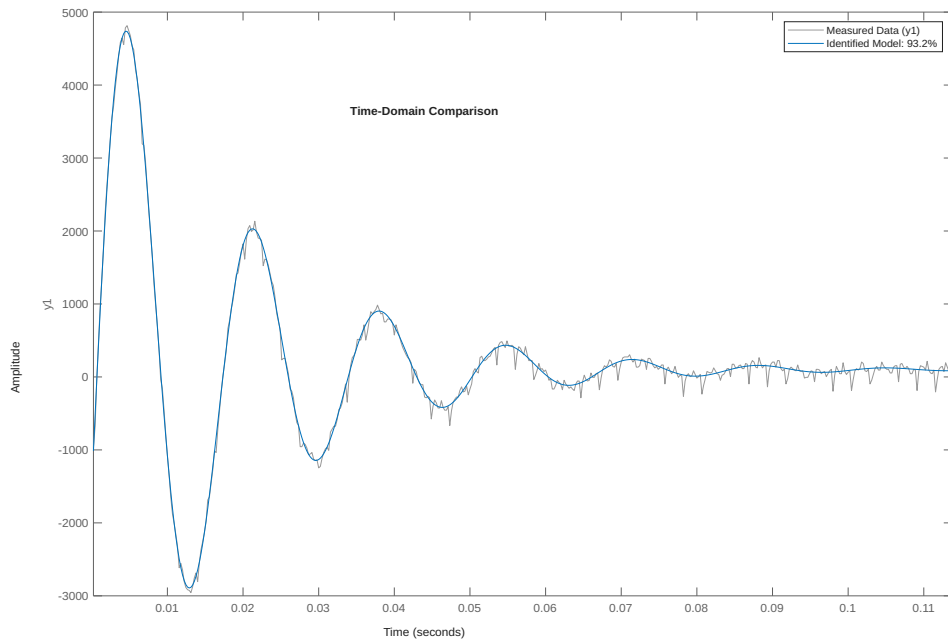


Figure 1.3: Step response of the identified model over the recorded data.

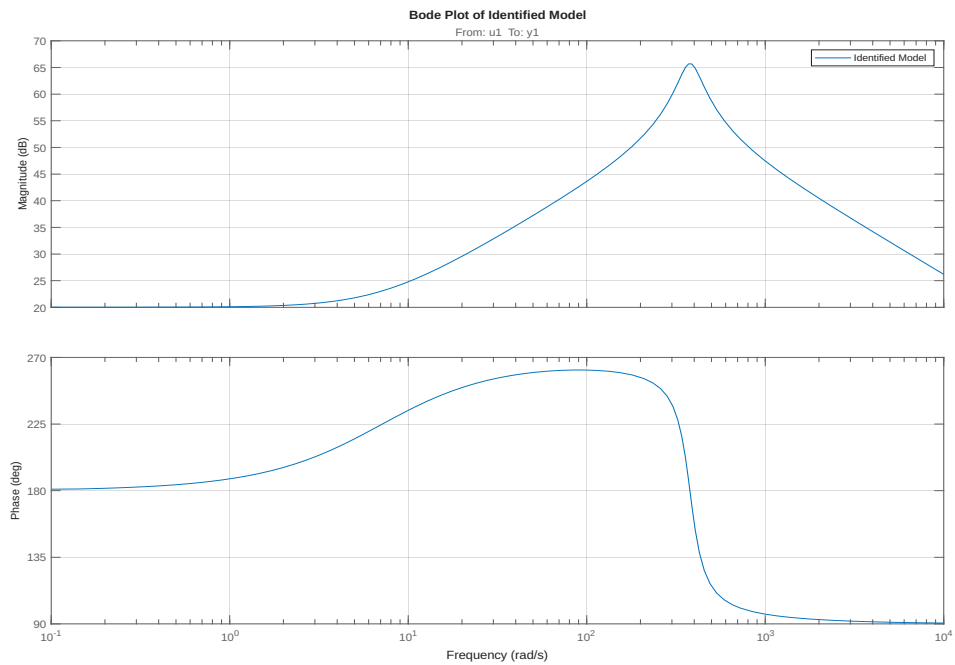


Figure 1.4: Bode plot of the identified system.

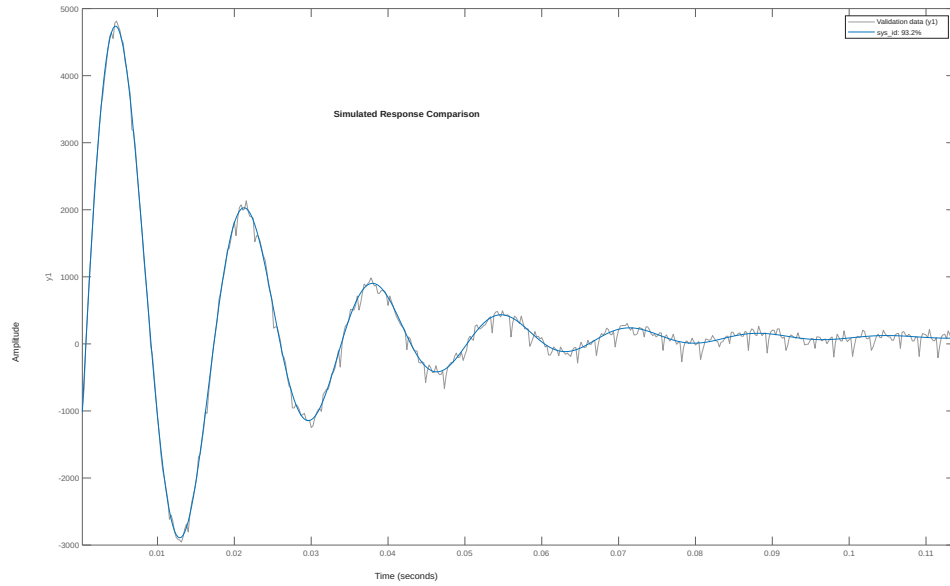


Figure 1.5: Step response of the identified corrected model over the recorded data.

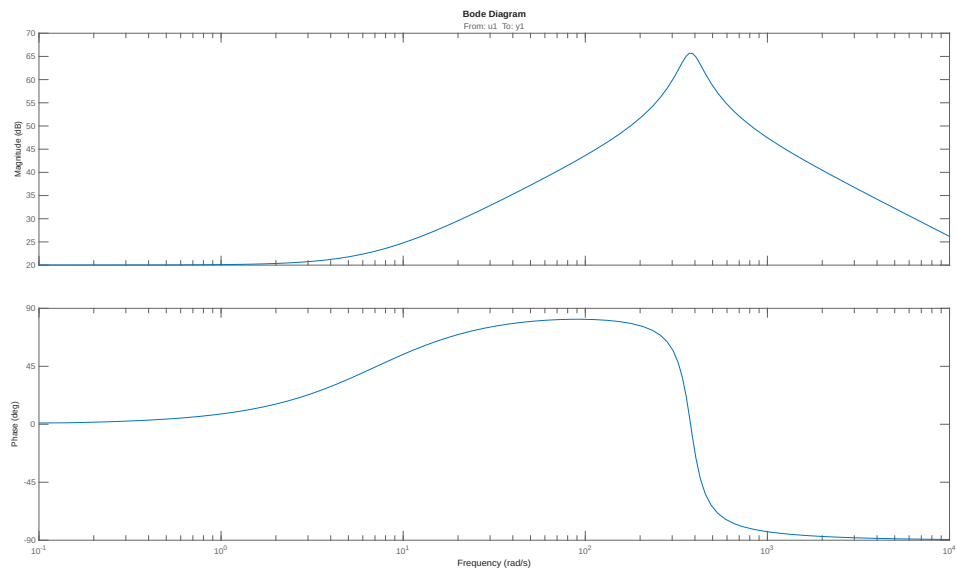


Figure 1.6: Bode plot of the identified corrected system.

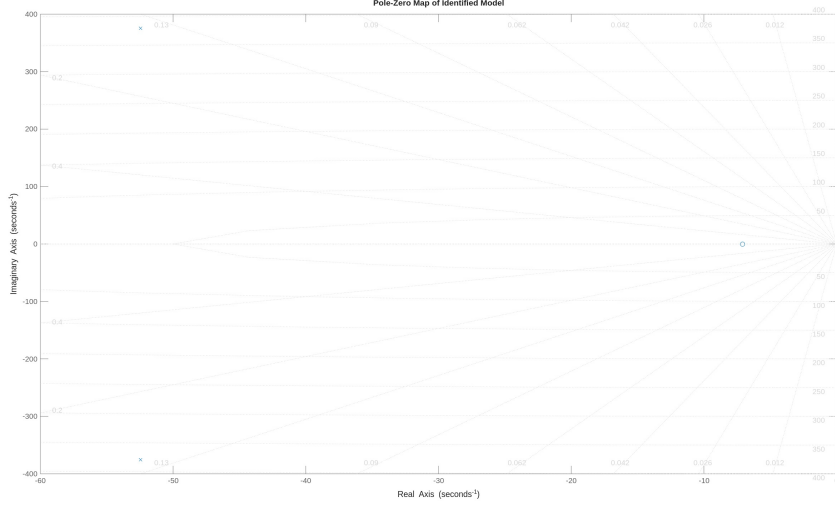


Figure 1.7: Pole-Zero-Plot of the identified corrected system.

system. However, as these were the actual values that we used during the lab, we chose not to change them.

1.10 I Controller Design

To design a pure integral controller such that the open-loop system has unit gain at a desired crossover frequency ω_c , the following steps are performed:

$$\omega_c = 2\pi f_c \quad (1.13)$$

We first evaluate the magnitude of the plant $P(s)$ at $s = j\omega_c$:

$$|P(j\omega_c)| = |P(s)|_{s=j\omega_c} \quad (1.14)$$

Then, to achieve a gain of 1 (0 dB) at the crossover frequency, the integral gain K_i is chosen as:

$$K_i = \frac{\omega_c}{|P(j\omega_c)|} \quad (1.15)$$

This yields the integral controller:

$$C(s) = \frac{K_i}{s} \quad (1.16)$$

The open-loop transfer function becomes:

$$L(s) = C(s)P(s) = \frac{K_i}{s}P(s) \quad (1.17)$$

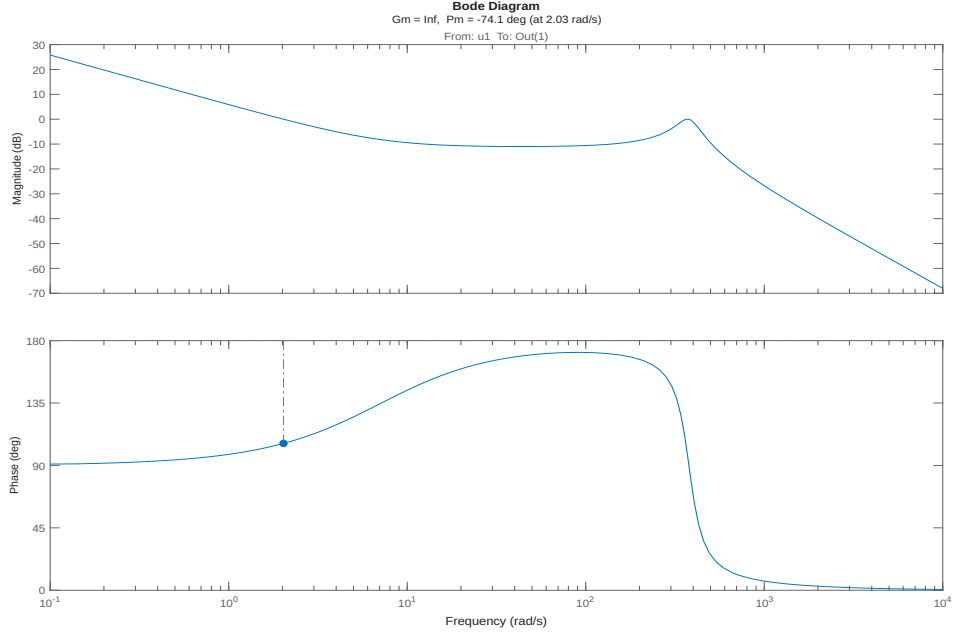


Figure 1.8: Bode plot of $L = C(s) * P(s)$.

This ensures:

$$|L(j\omega_c)| = 1 \quad (1.18)$$

i.e., the gain crossover frequency is exactly ω_c , as desired.

Using our estimated model $P(s)$, we can find the integral gain K_i such that our system has its magnitude zero crossing at $f_c = 60$ Hz. With this strategy, we estimated

$$K_i = 0.1944. \quad (1.19)$$

Verifying with the bode plot in 1.8 confirms our chosen K_i . Discretizing in the last step, yields the discrete pure integral controller

$$C(z)^i = \frac{4.3199 \cdot 10^{-5}}{1 - z^{-1}}, \quad (1.20)$$

with the discrete pure integral controller gain

$$K_{i,d} = 4.3199 \times 10^{-5}. \quad (1.21)$$

1.11 Discrete C Implementation

The estimated discrete pure integral controller is now implemented in the C-code given in 1.1 and synthesized for the FPGA.

```

1 int reference = 0; // our reference is zero -> we are
  just stabilizing the plant
2 int t_phi = MirrorGetPhase(); // t_phi is what we
  measure at the output of the plant
3
4 int delta_t_phi = mirror->t_phi_lin - t_phi;
5 double error = delta_t_phi;
6
7 float Kp = 0;
8 float Ki = 4.3199e-5f;
9
10 double delta_f = mirror->integrator + mirror->Kp * error;
11 mirror->integrator = mirror->integrator + mirror->Ki *
  error;
12
13 double f = delta_f + mirror->f_lin;
14
15 MirrorSetFrequency(f);

```

Listing 1.1: Integral Controller C-Code

1.12 Disturbance Rejection

To evaluate our implemented discrete I-Controller, an artificial disturbance is applied to the system while we measure the phase error in time $\Delta t\phi$. The step was also applied to the simulated closed loop consisting of the estimated integral controller and identified system. The comparison between the measured and simulated response can be seen in 1.9. Both signals are quite similar, confirming both our system and designed controller. However, a slight deviation, both in phase and amplitude has to be acknowledged. These differences can be ascribed to the model uncertainties of the identified linear model.

1.13 PI Controller Design

In this task, we designed a PI controller, again with the requirements of unit gain at ω_c and a phase margin of 70° at a crossover frequency of $f_c = 200$ Hz. The PI controller has the general form

$$C(s) = K_p \left(1 + \frac{1}{T_i s} \right), \quad (1.22)$$

The corresponding angular frequency is, again, computed as

$$\omega_c = 2\pi f_c. \quad (1.23)$$

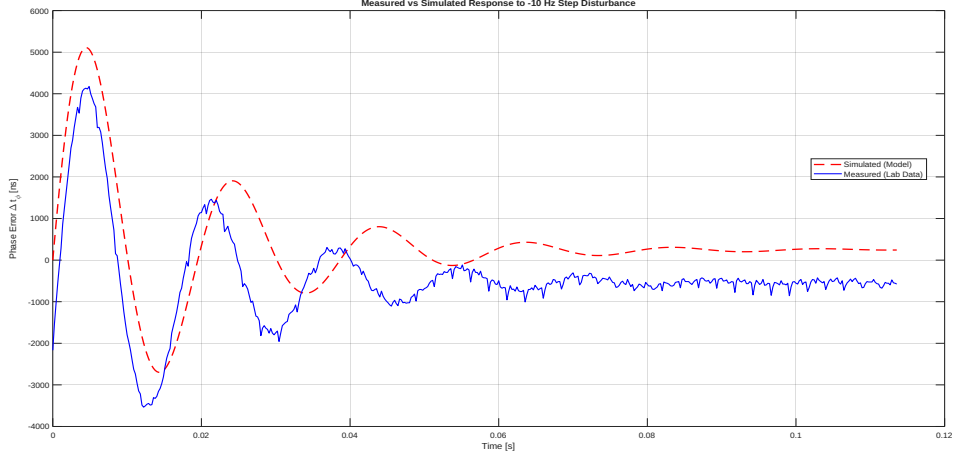


Figure 1.9: System response to a 10 Hz input frequency step. Simulated and measured.

We first evaluate the frequency of $P(s)$ at ω_c and calculate the gain and phase at this frequency with

$$|P(j\omega_c)|, \quad \angle P(j\omega_c). \quad (1.24)$$

The current phase margin is calculated as

$$\varphi_{\text{current}} = 180^\circ + \angle P(j\omega_c). \quad (1.25)$$

To achieve the desired phase margin of at least 70° our controller must add

$$\Delta\varphi = \varphi_{\text{desired}} - \varphi_{\text{current}}. \quad (1.26)$$

to the open loop consisting of controller and system $L(s) = C(s)P(s)$.

To achieve this, the integral time constant T_i of the PI controller is chosen such that it adds the necessary phase lead at the crossover frequency. Therefore:

$$T_i = \frac{\tan\left(\frac{\pi}{2} + \Delta\varphi_{\text{rad}}\right)}{\omega_c}, \quad (1.27)$$

where $\Delta\varphi_{\text{rad}}$ is the required phase lead in radians. With T_i fixed, the proportional gain K_p is still to be determined. We can choose K_i such, that the magnitude of the open-loop transfer function at ω_c is at unity at the ω_c . This is achieved by choosing

$$K_p = \frac{1}{|P(j\omega_c)| \cdot \sqrt{1 + \left(\frac{1}{\omega_c T_i}\right)^2}}. \quad (1.28)$$

With this method, we calculated

$$T_i = 0.0017124, \quad (1.29)$$

$$K_p = 0.0051082. \quad (1.30)$$

Validating the chosen values for K_i and T_i is done via the open loop bode plot given in ??.

Converting to a discrete PI controller

$$C(z) = K_{p,d} + \frac{K_{i,d}}{z - 1} \quad (1.31)$$

the discrete integral and proportional gains calculate to

$$K_{p,d} = K_p = 0.0051082, \quad (1.32)$$

$$K_{i,d} = \frac{K_p T_s}{T_i} = \frac{0.0051082 \cdot \frac{1}{4500}}{0.0017124} = 0.000629. \quad (1.33)$$

1.13.1 Disturbance Rejection PI Control

For evaluation we again test the system with the same artificial disturbance, we already used for the evaluation of the I controller. Both the simulated and measured responses to the disturbance can be seen in 1.10. It is noted, that for this comparison the corrected identified model of the system was used, as for the initial one, the simulated closed loop was not stable. With this model then, the simulated and the measured response to the disturbance match very well. Compared with the disturbance rejection from the pure integral controller 1.9, the PI controller clearly returns the system to zero much faster and without significant overshoot, while the pure integral controller does oscillates quite significantly, before returning to zero again.

1.14 Control Loop Affects on Timing Jitter

Measuring the jitter with MATLAB's *std* function, the timing jitter of the open loop, with pure integral control and with the PI controller is measured:

$$\sigma_{\Delta t} = 18.8504 \text{ ns}, \quad (1.34)$$

$$\sigma_{\Delta t}^{\text{I-Control}} = 83.0761 \text{ ns}, \quad (1.35)$$

$$\sigma_{\Delta t}^{\text{PI-Control}} = 79.8076 \text{ ns}. \quad (1.36)$$

$\sigma_{\Delta t}^{\text{I-Control}}$ is likely slightly overestimated, as it was calculated from the disturbance rejection data, where the integral controller had not fully rejected the disturbance by the end of the measurement.

Still, the verdict is clear. Both controllers substantially increase the timing jitter. This is somewhat expected behavior, as the already present jitter gets amplified by the control loops.

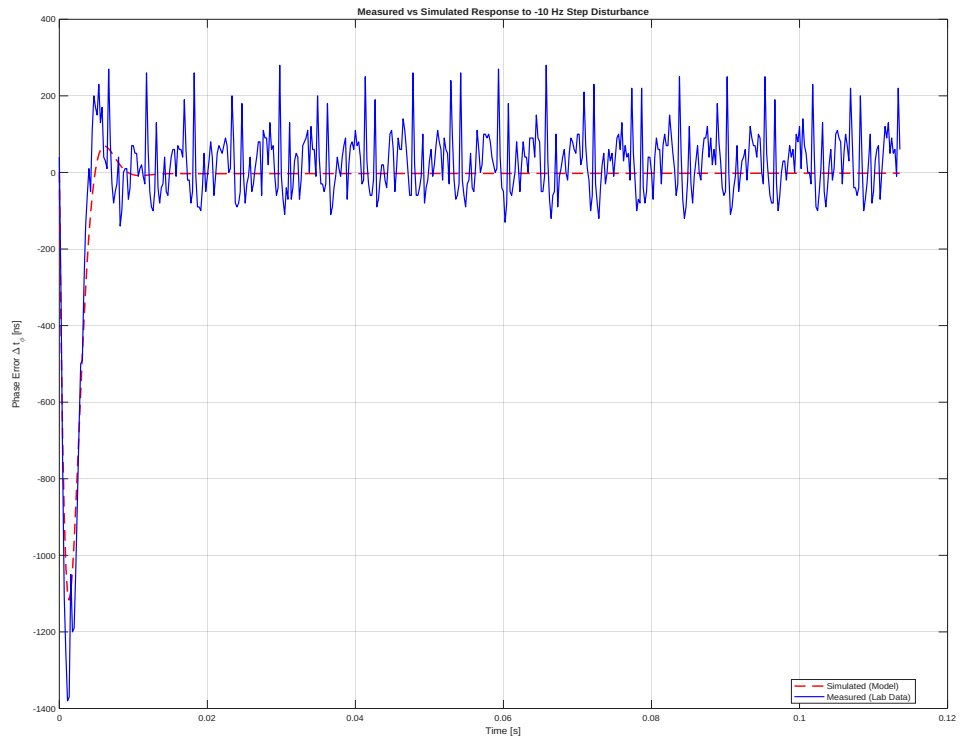


Figure 1.10: System response of PI controller closed loop to a 10 Hz input frequency step. Simulated and measured.

Report Questions

2.1 Backbone Curve

The backbone curve, here shown in 2.1, represents the frequency response of the system given no activation torque $\tau_{\text{comb}} = 0$. As stated in the laboratory instructions, when V_{act} is nonzero, the resulting torque τ_{comb} is always directed such that it pulls the mirror back toward the center position. Basically acting like an additional spring, effectively increasing the system's overall stiffness. As a consequence, the resonance frequency increases, which explains the shift to the right in the frequency response compared to the purely mechanical backbone curve.

2.2 Higher Gain for Integral Control or Differential Action?

Just increasing the gain of the integral controller will not automatically bring better results. As seen in Figure 1.9, the overshoot in the disturbance rejection is already quite significant. With higher gain, this issue would just become even more prominent.

Adding differential action to our controller is also challenging, as the timing jitter is already quite high. A differential controller would make this exponentially worse, as it is highly sensitive to high-frequency noise.

2.3 Comparison to a Driven Harmonic Oscillator

The driving torque of the MEMS mirror system

$$\tau_{\text{comb}} \propto \frac{dC}{d\theta} V_{\text{act}}^2 \quad (2.1)$$

is both dependent on the mirror angle θ and nonlinearly dependent on the driving voltage V_{act} . For a driven harmonic oscillator, the driving torque is generally sinusoidal and independent of θ .

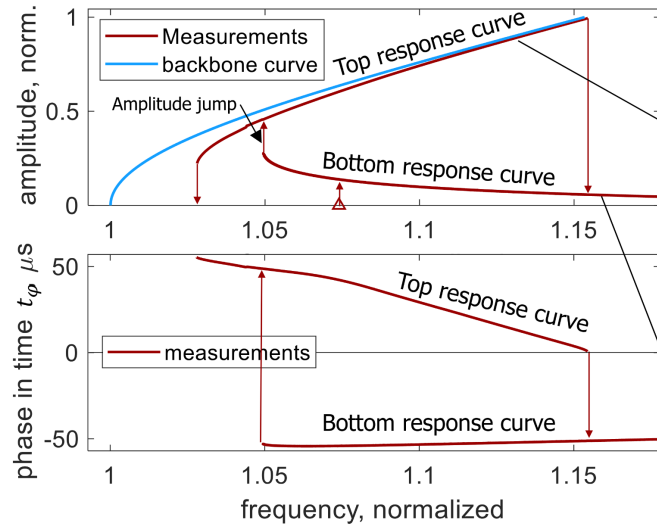


Figure 2.1: Normalized response curves of the mirror from the laboratory instructions.

A linear oscillator would exhibit a symmetric resonance peak, and importantly would not exhibit hysteresis.

The backbone curve of a linear oscillator is flat, as the resonance frequency

$$f(\Theta) = \frac{1}{2\pi} \sqrt{\frac{k}{J}} \quad (2.2)$$

is independent of the amplitude and constant.